

LOSO Experiment A1_Istm

LSTM variants only, A1 config (incl. Double_Nothing)

Generated 2026-05-11 17:27:08

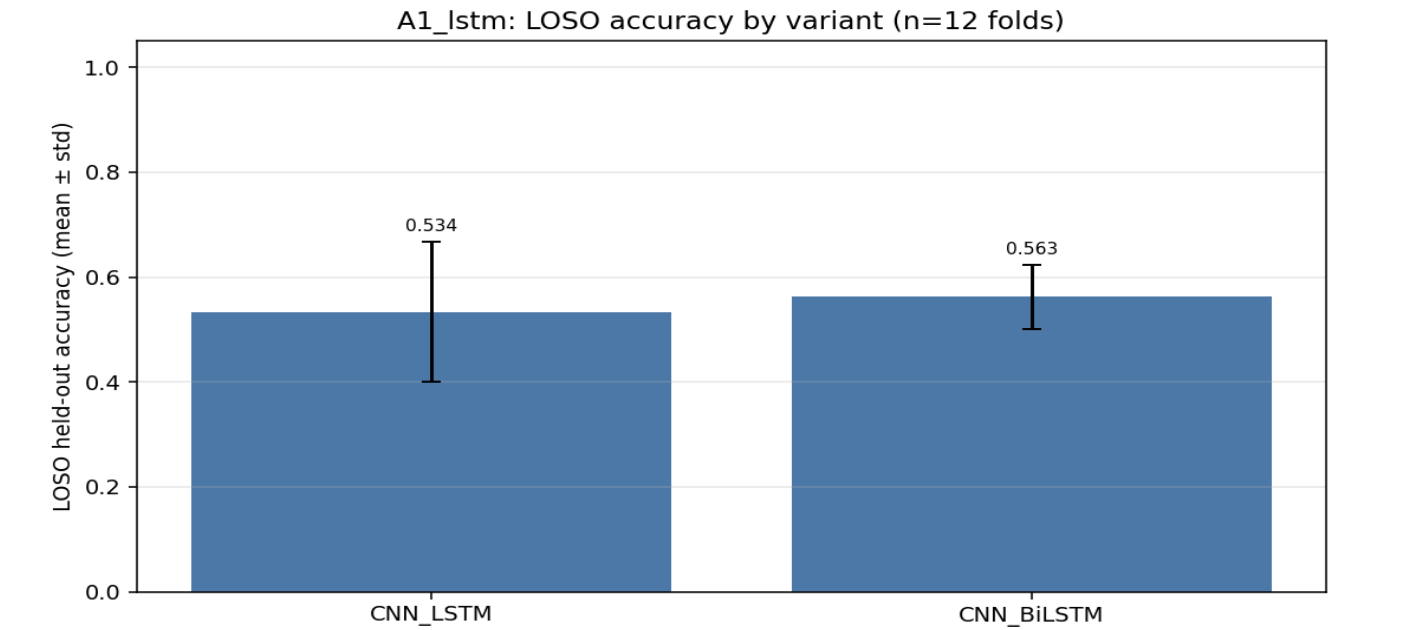
Experiment configuration

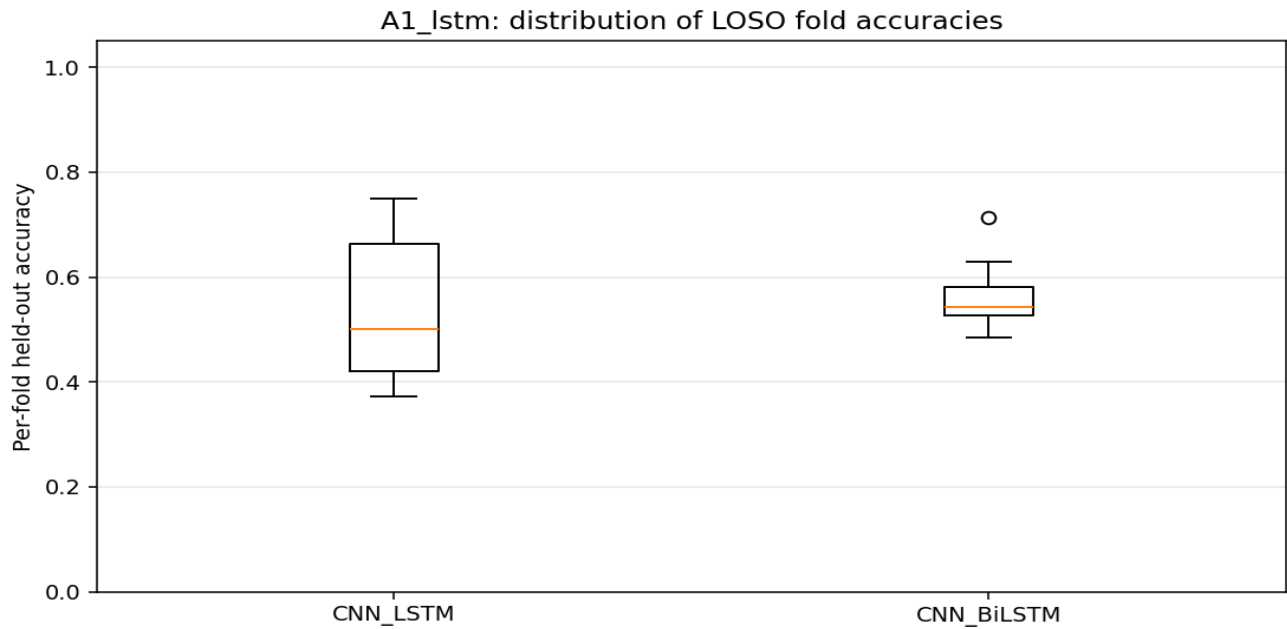
Class set	7 classes including Double_Nothing
Classes	Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR, accel
Hands	left, right
Sensor channels	176
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

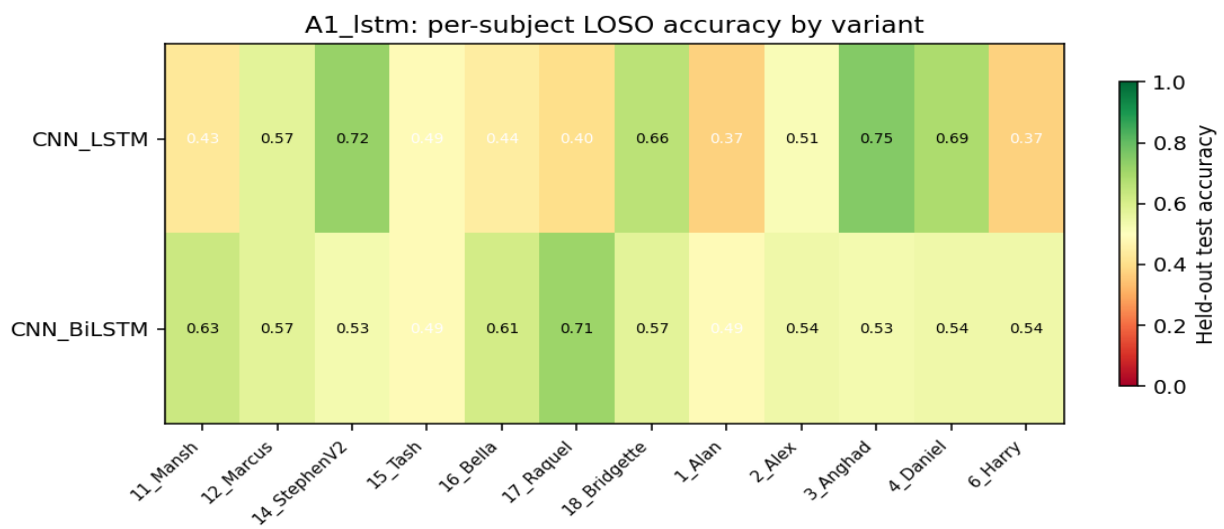
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	68,455	0.5335	0.1336	0.3714	0.7500	0.6924
CNN_BiLSTM	105,575	0.5627	0.0614	0.4857	0.7143	0.7476

Best variant by LOSO mean: **CNN_BiLSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	388	0.429	0.629
12_Marcus	388	0.571	0.571
14_StephenV2	387	0.722	0.528
15_Tash	388	0.486	0.486
16_Bella	387	0.444	0.611
17_Raquel	388	0.400	0.714
18_Bridgette	388	0.657	0.571
1_Alan	388	0.371	0.486
2_Alex	388	0.514	0.543
3_Anghad	387	0.750	0.528
4_Daniel	388	0.686	0.543
6_Harry	388	0.371	0.543

Variant: CNN_LSTM

Params: 68,455

LOSO mean accuracy: 0.5335 ± 0.1336 (min 0.3714, max 0.7500)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	0/5	5/5	0/5	5/5	0/5	0.429
12_Marcus	5/5	0/5	5/5	5/5	0/5	5/5	0/5	0.571
14_StephenV2	2/5	3/6	5/5	5/5	1/5	5/5	5/5	0.722
15_Tash	5/5	0/5	5/5	5/5	0/5	2/5	0/5	0.486
16_Bella	5/5	1/6	5/5	0/5	0/5	5/5	0/5	0.444
17_Raquel	0/5	0/5	5/5	5/5	0/5	4/5	0/5	0.400
18_Bridgette	5/5	3/5	5/5	5/5	0/5	5/5	0/5	0.657
1_Alan	5/5	3/5	0/5	4/5	1/5	0/5	0/5	0.371
2_Alex	0/5	3/5	4/5	5/5	0/5	1/5	5/5	0.514
3_Anghad	5/5	2/5	5/5	4/5	0/5	6/6	5/5	0.750
4_Daniel	5/5	2/5	5/5	2/5	0/5	5/5	5/5	0.686
6_Harry	0/5	0/5	5/5	0/5	3/5	0/5	5/5	0.371
MEAN	0.700	0.272	0.817	0.750	0.083	0.700	0.417	0.534

Variant: CNN_BiLSTM

Params: 105,575

LOSO mean accuracy: 0.5627 ± 0.0614 (min 0.4857, max 0.7143)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	4/5	0/5	5/5	3/5	5/5	0/5	0.629
12_Marcus	4/5	0/5	4/5	5/5	2/5	5/5	0/5	0.571
14_StephenV2	5/5	1/6	5/5	3/5	0/5	0/5	5/5	0.528
15_Tash	5/5	0/5	5/5	5/5	0/5	2/5	0/5	0.486
16_Bella	5/5	2/6	5/5	1/5	0/5	5/5	4/5	0.611
17_Raquel	5/5	5/5	5/5	5/5	0/5	5/5	0/5	0.714
18_Bridgette	5/5	0/5	5/5	5/5	0/5	5/5	0/5	0.571
1_Alan	5/5	2/5	5/5	0/5	0/5	5/5	0/5	0.486
2_Alex	5/5	1/5	3/5	5/5	0/5	0/5	5/5	0.543
3_Anghad	3/5	0/5	3/5	5/5	0/5	6/6	2/5	0.528
4_Daniel	5/5	0/5	5/5	5/5	0/5	0/5	4/5	0.543
6_Harry	3/5	0/5	5/5	5/5	2/5	4/5	0/5	0.543
MEAN	0.917	0.242	0.833	0.817	0.117	0.683	0.333	0.563